Uncertainty-Aware G/G/k Autoscaling Pipeline

RF service-time prediction with error propagation · P95 SLA on end-to-end latency · minimum total waiting time

1. Workload Data Collection

Request logs & system state

- Arrival timestamps
- Job features x = (CPU, RAM, type)
- Ingress buffer $B(t)$, queue $Q(t)$
- Active + booting workers $k(t)$
- Completed jobs: runtime S, storage T_{st}

2. Arrival Characterization

Estimate from recent window:

- Arrival rate: $\lambda = 1/E[A]$
- Arrival variability:

$$C_a^2 = \frac{\mathrm{Var}(A)}{(E[A])^2}$$

- Optional: forecast λ at $t + T_{scale}$

NOVEL · N1

3. RF Service-Time Oracle

Per job, at admission time:

- Mean runtime: $\hat{S}(x)$
- Error variance: $\hat{\sigma}^2(x)$
  (quantile RF / OOB-residual RF)
- P95 runtime: $\hat{S}_{95}(x)$
- Validated on held-out jobs

NOVEL · N1

4. Corrected Service Moments

σ^2

- Service rate: $\mu = 1/E[\hat{S}]$

Law of total variance:

$$\mathrm{Var}(S) = \mathrm{Var}(\hat{S}) + E[\hat{\sigma}^2(x)]$$

$$C_s^2 = \frac{\mathrm{Var}(S)}{(E[S])^2}$$

Naive plug-in drops $E[\hat{\sigma}^2] \rightarrow C_s^2$ too low

NOVEL · N3

5. Boot-Delay Lookahead

- New workers ready at $t + T_{scale}$
- k_{act} = active workers (booting do not serve)

Projected backlog at boot completion:

$$\tilde{Q} = Q(t) + B(t)$$

$$+ \max(0,\ \lambda - k_{act}\,\mu)\,T_{scale}$$

6. Erlang-C (M/M/k Reference)

- $a = \lambda/\mu, \quad \rho = a/k, \quad \rho < 1$

$$P_W = \frac{\frac{a^k}{k!\,(1-\rho)}}{\sum_{n=0}^{k-1} \frac{a^n}{n!} + \frac{a^k}{k!\,(1-\rho)}}$$

Mean wait for M/M/k:

$$W_q^{M/M/k} = \frac{P_W}{k\mu - \lambda}$$

7. Variability Correction

Allen–Cunneen factor (corrected C_s^2):

$$V = \frac{C_a^2 + C_s^2}{2}$$

Estimated waiting time for G/G/k:

$$W_q^{G/G/k} \approx V \cdot \frac{P_W}{k\mu - \lambda}$$

V = 1 recovers M/M/k (unit test)

NOVEL · N2

8. SLA Evaluation on W_total

$$W_{total} = W_B + W_q + S + T_{st}$$

Per-request tail, averaged over RF runtimes:

$$P(W_{total} > T_{SLA}) \approx E_S\left[P_W e^{-(k\mu - \lambda)\tau/V}\right]$$

$$\tau = T_{SLA} - S - T_{st} - \tilde{Q}/(k\mu)$$

Feasible if $\leq \varepsilon$ (P95 SLA); replaces $\alpha = 0.8$

9. Capacity Decision: Minimise Total Waiting

$$k^* = \arg\min_k\ \lambda\,E[W_B + W_q](k) + c\,k \quad \text{s. t.} \quad P(W_{total} > T_{SLA}) \leq \varepsilon,\ \rho < 1$$

compare k^* with k

- $k^* > k$: scale up now, $k \leftarrow k^*$
- $k^* = k$: hold
- $k^* < k$ for N intervals: $k \leftarrow \max(k^*, k_{min})$

10. Apply Scaling Decision

- Provision / terminate VMs
- Boot delay T_{scale} = 10 s; count booting VMs
- Enforce $k_{min} \leq k \leq k_{max}$
- Cooldown before any scale-down

11. Measure & Recalibrate

- W_B, W_q, S, T_{st}; total waiting; SLA viol.
- VM-hours, buffer overflow rate
- RF interval coverage; predicted vs observed W_q
- Update λ, C_a^2; refit RF on drift

Repeat every control interval (T_{ctrl} = 4 s)

Goal: minimise the total waiting time of all requests within a P95 SLA and VM budget

$$\min\ \int(\lambda\,E[W_B + W_q] + c\,k(t))dt \quad \text{s. t.} \quad P(W_{total} > T_{SLA}) \leq \varepsilon$$

Novelty of the proposed model

N1 ML prediction error propagated into C_s^2 (steps 3–4)
N2 P95 SLA on W_{total} (step 8) N3 boot-delay lookahead (step 5)